THE MATHEMATICS OF OMI

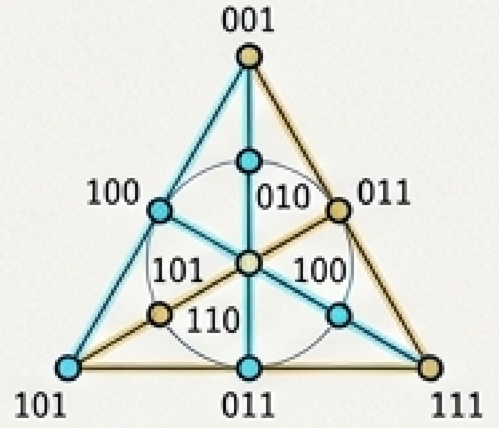
COMBINATORICS, PROJECTIVE GEOMETRY, AND THE LOGICAL PHYSICS OF THE OMICRON OBJECT MODEL

PROJECTIVE INCIDENCE STRUCTURES

Axiom P1: For any two distinct points, there is exactly one line incident with both.

Axiom P2: For any two distinct lines, there is exactly one point incident with both.

Axiom P3: There exist four points, no three of which are collinear.

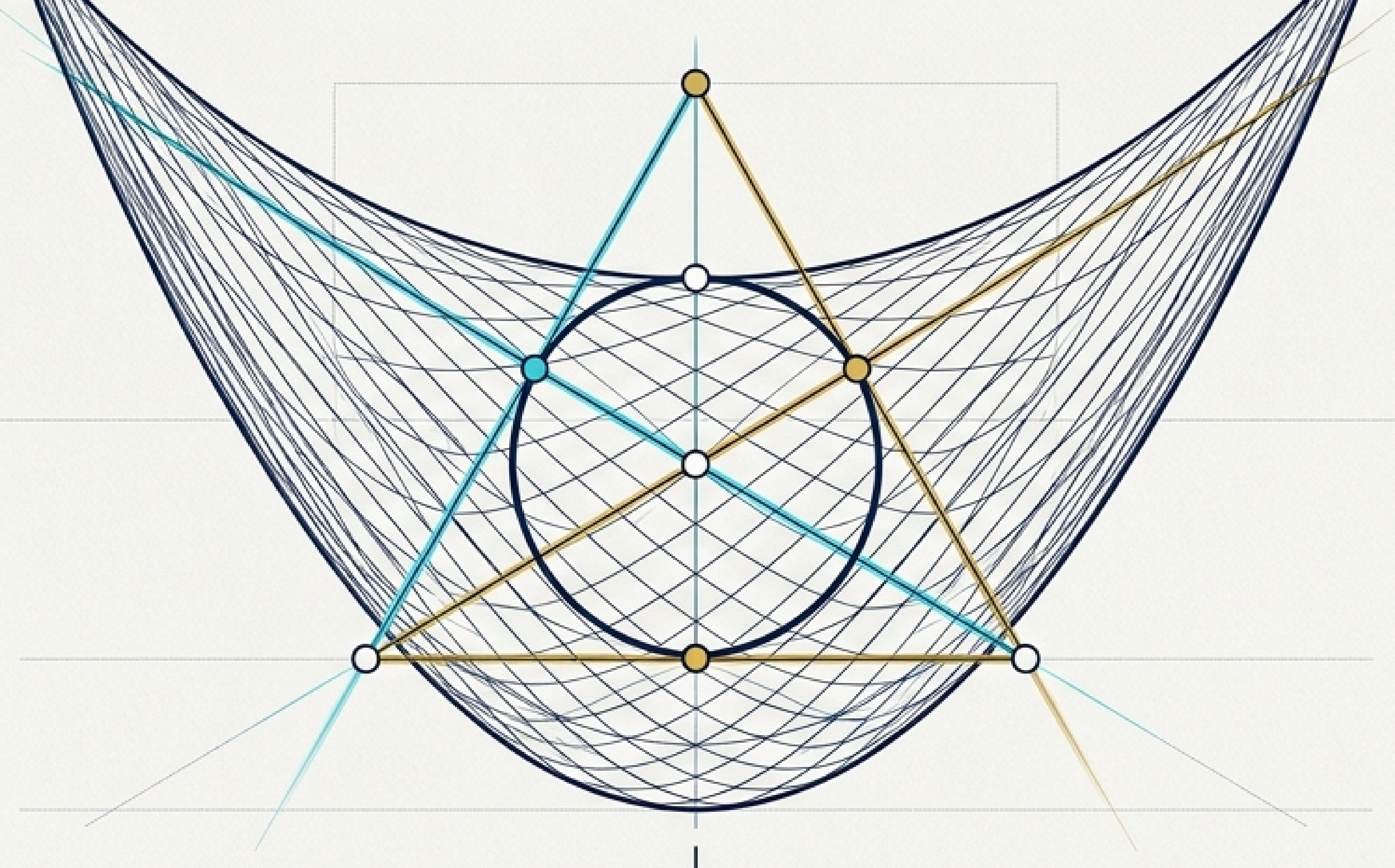


Fano Plane F_7: (P, L, I)

$|P| = 7$, $|L| = 7$

$\mathrm{Aut}(F_7) \cong \mathrm{PGL}(3,2) \cong \mathrm{PSL}(2,7)$

Order = 168



THE ONI FORMALISM: The intersection of the continuous elliptic paraboloid manifold with the discrete Fano plane defines the quantized state space of the Omicron Object. The resulting configuration encodes the logical physics, mapping combinatorial constraints to dynamical energy surfaces.

ELLIPTIC PARABOLOID & OMICRON DYNAMICS

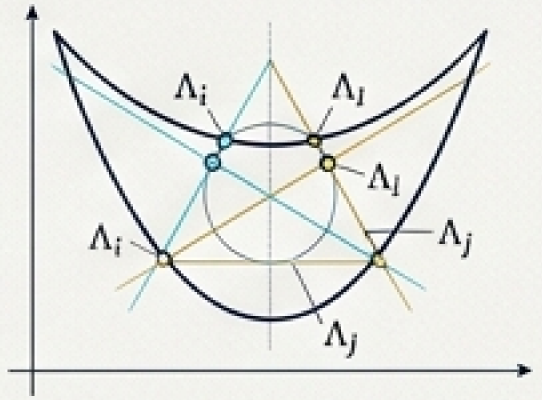
Equation in Cartesian Coordinates:

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Parametrization:

$$r(u, v) = (a\,u\cos(v), b\,u\sin(v), u^2)$$

Intersection Locus $\Lambda = P \cap F_7$



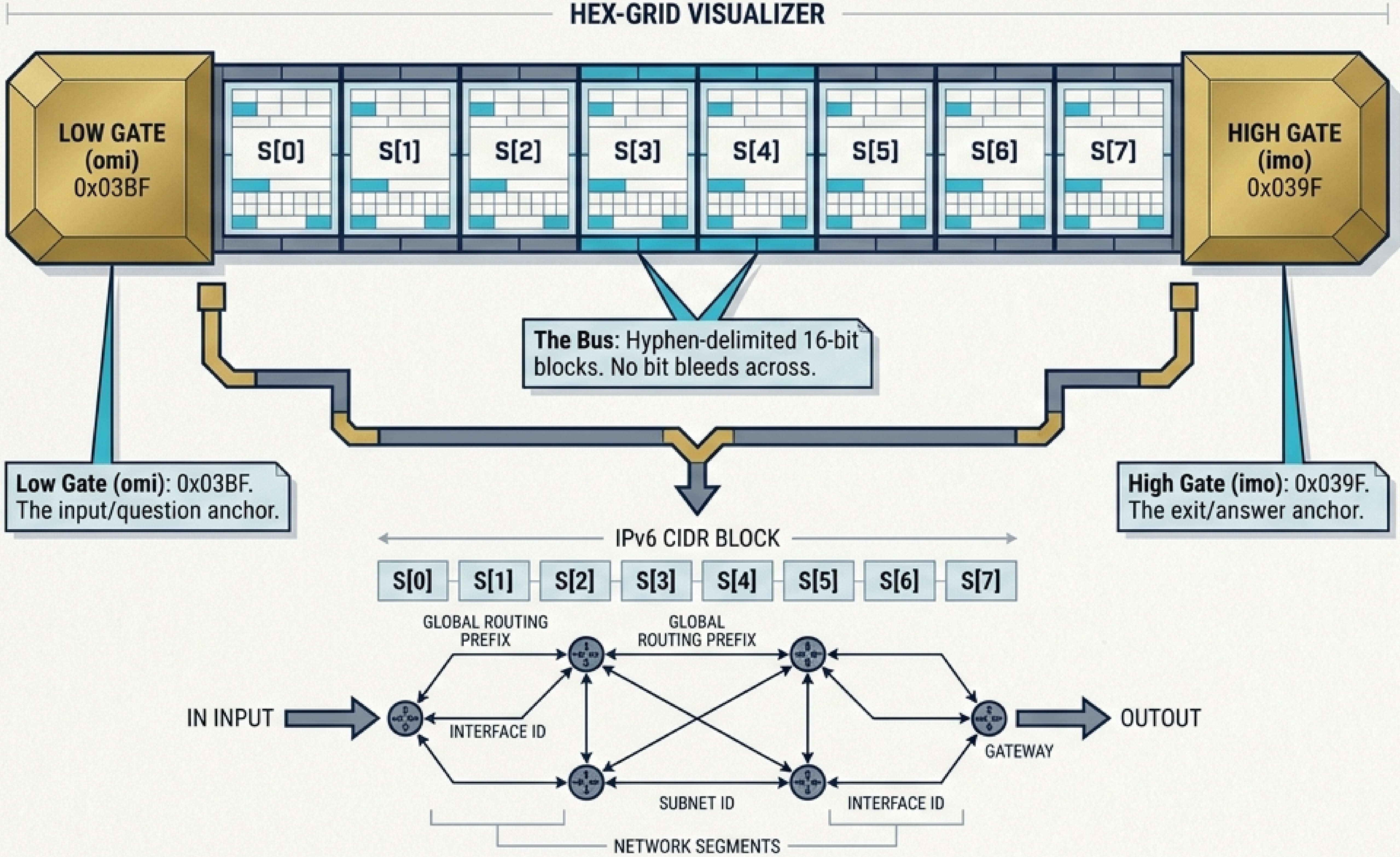
Constraint Matrix C_Ω:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\det(C_\Omega) \neq 0$ (over GF(2))

State Vector $\Psi \in H_\Omega, H_\Omega \cong \mathbb{C}^7$

THE ATOMIC FRAME: BOUNDED BY THE PALINDROME

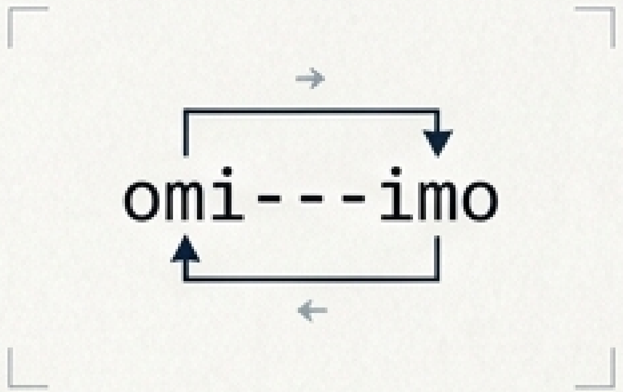


CORE SUMMARY

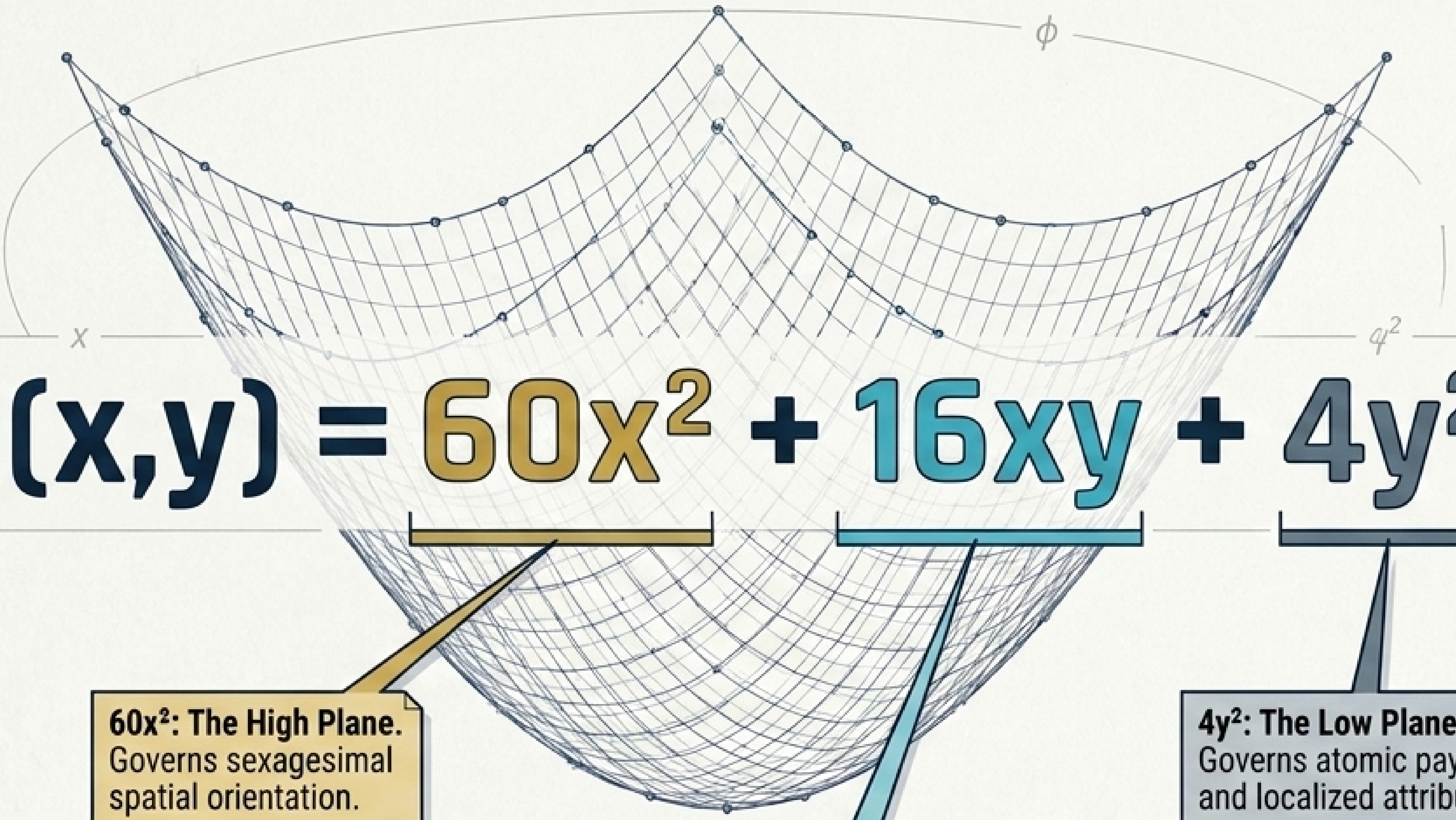
- Every valid transmission is a mathematically bounded 2^{10} (1024-bit) instruction word framed within exactly 128 hexadecimal characters (2^7).

$$N_{total} = 2^{10} = 1024 \text{ bits}$$
$$L_{hex} = 2^7 = 128 \text{ chars}$$

- The palindrome **omi---imo** functions as an unbreakable structural boundary. Inverting the sequence flips the processing matrix perfectly right-to-left.



THE SPATIAL EQUATION: BOUNDING THE EXECUTION SURFACE



$$Q(x,y) = 60x^2 + 16xy + 4y^2$$

$60x^2$: The High Plane. Governs sexagesimal spatial orientation.

$16xy$: The Cross-Term Bridge. The active execution matrix mapping structure to data.

$4y^2$: The Low Plane. Governs atomic payload and localized attributes.

MATHEMATICAL GUARANTEE

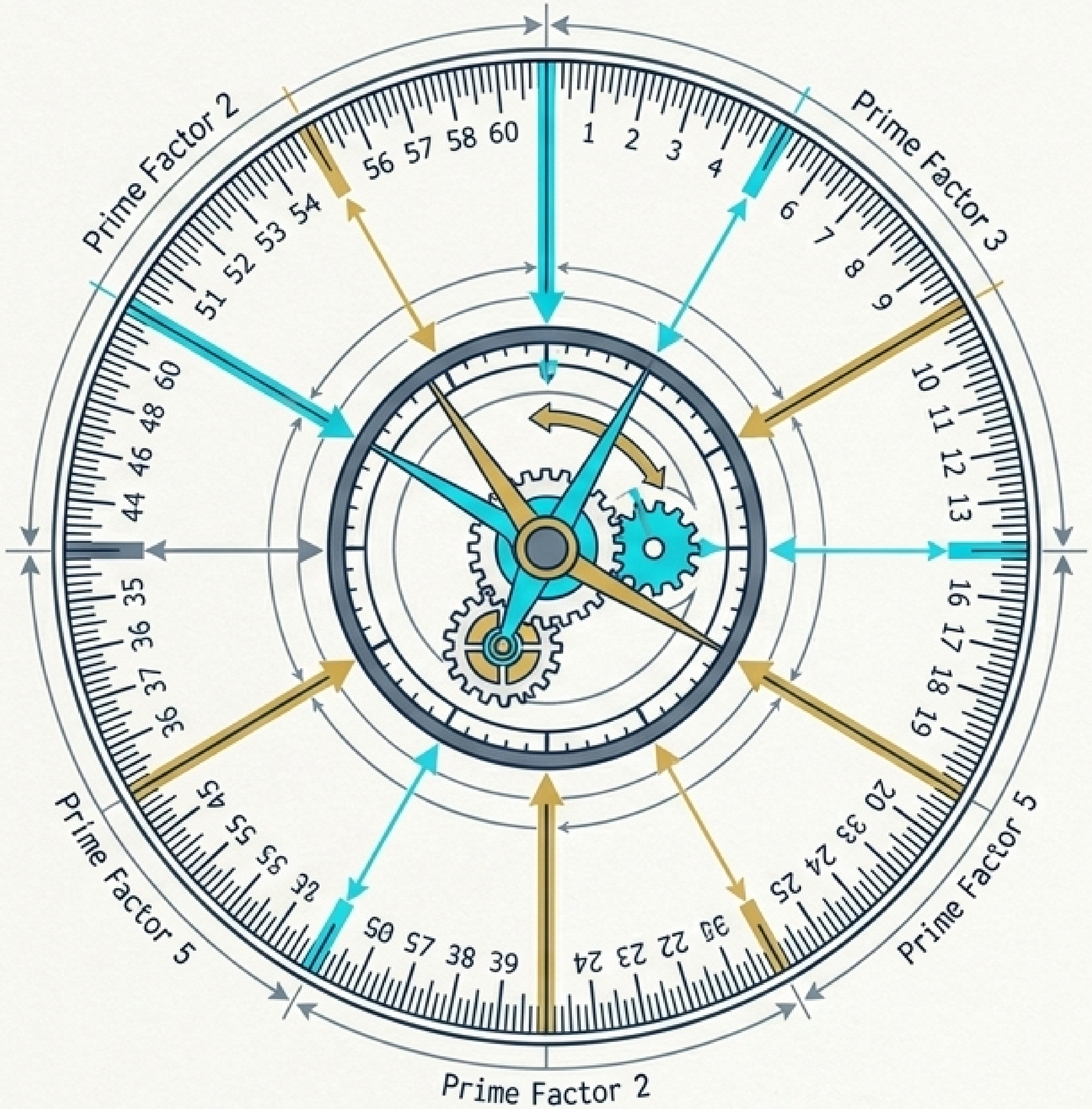
The positive-definite nature of the matrix (det > 0) guarantees a convex, bounded elliptic paraboloid surface.

PHYSICAL MEANING

Valid coordinate pairs form a closed execution space with absolutely no saddle points for state to leak.

omi---imo

SEXAGESIMAL TOPOLOGY: THE ERADICATION OF TRUNCATION



The coefficient 60 is a superior highly composite number. Its divisibility maps a unit perfectly without floating-point errors.

Divisors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
1/2 → Exact Sexagesimal: 0;30
1/3 → Exact Sexagesimal: 0;20
1/4 → Exact Sexagesimal: 0;15
1/5 → Exact Sexagesimal: 0;12
1/6 → Exact Sexagesimal: 0;10

Structural Note: Because denominators with prime factors 2, 3, and 5 resolve exactly, subdividing Omi's spatial grid never produces a repeating decimal or a truncation error.

omi---imo

THE DELTA LAW: THE REVERSIBLE ENGINE

$$\delta_C(x) = \mathrm{rotl}(x,1) \oplus \mathrm{rotl}(x,3) \oplus \mathrm{rotr}(x,2) \oplus C$$

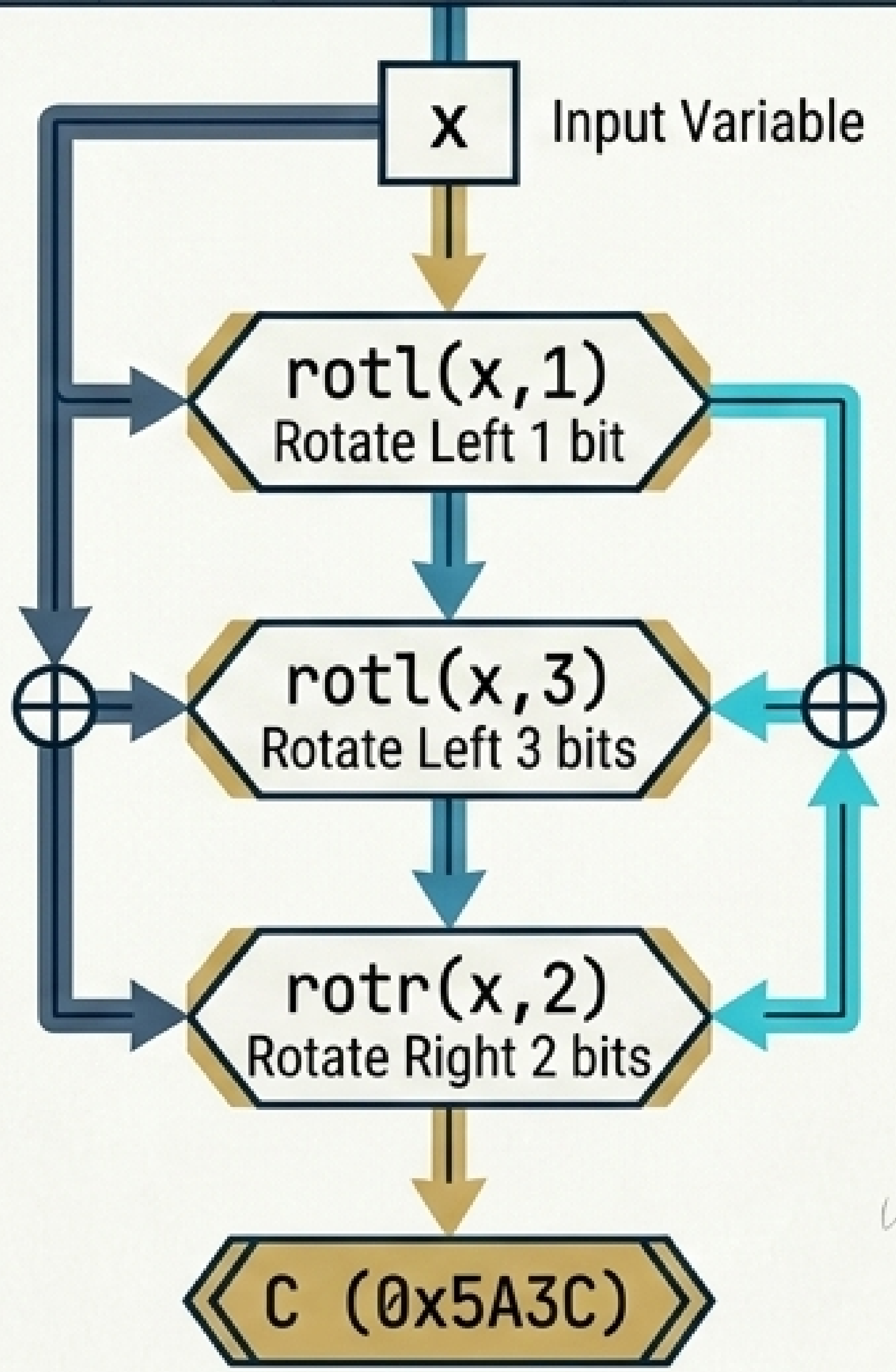
CONSTRAINT MATRIX (INPUTS)

Rotations Only: Zero bits are ever lost (no shifts).

XOR Logic: Fully reversible (XOR is its own inverse).

Mask to 16-bit: Keeps execution strictly bounded to uint16 space.

Constant C (0x5A3C): Mathematically breaks the zero fixed-point (0 → 0).

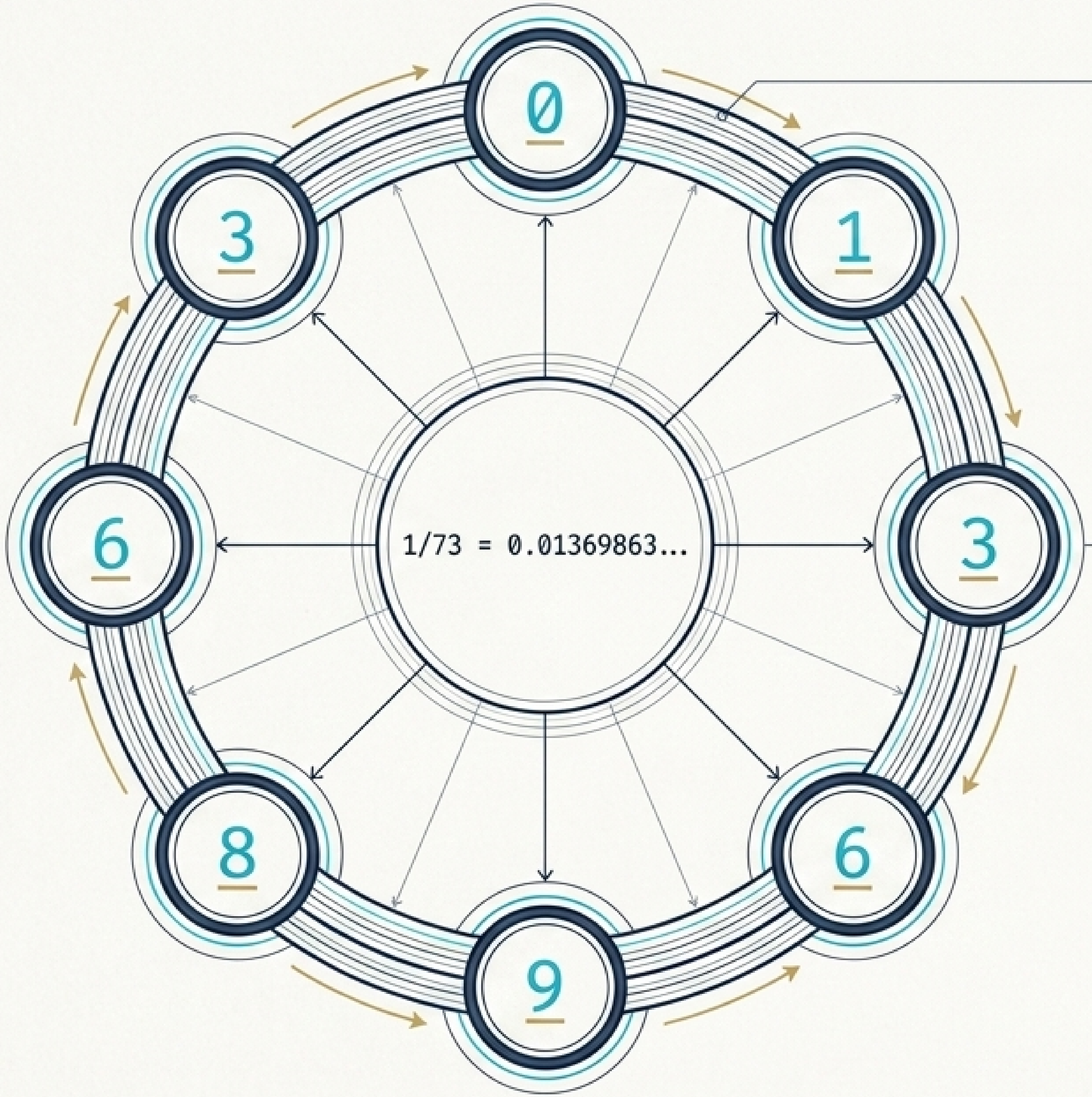


EMERGENT MATHEMATICAL PROPERTIES (OUTPUTS)

Produces a strict Period-8 closed orbit.

Intersects directly with finite projective geometry limits.

ORBITAL MECHANICS: THE PERIOD-8 CYCLE



THE PRIME 73 PHENOMENON

The reciprocal of Prime 73 perfectly generates the orbital weight sequence for Omi's state machine.

THE WEIGHT SEQUENCE ARRAY

[0, 1, 3, 6, 9, 8, 6, 3]

MASTER STRIDE DERIVATION

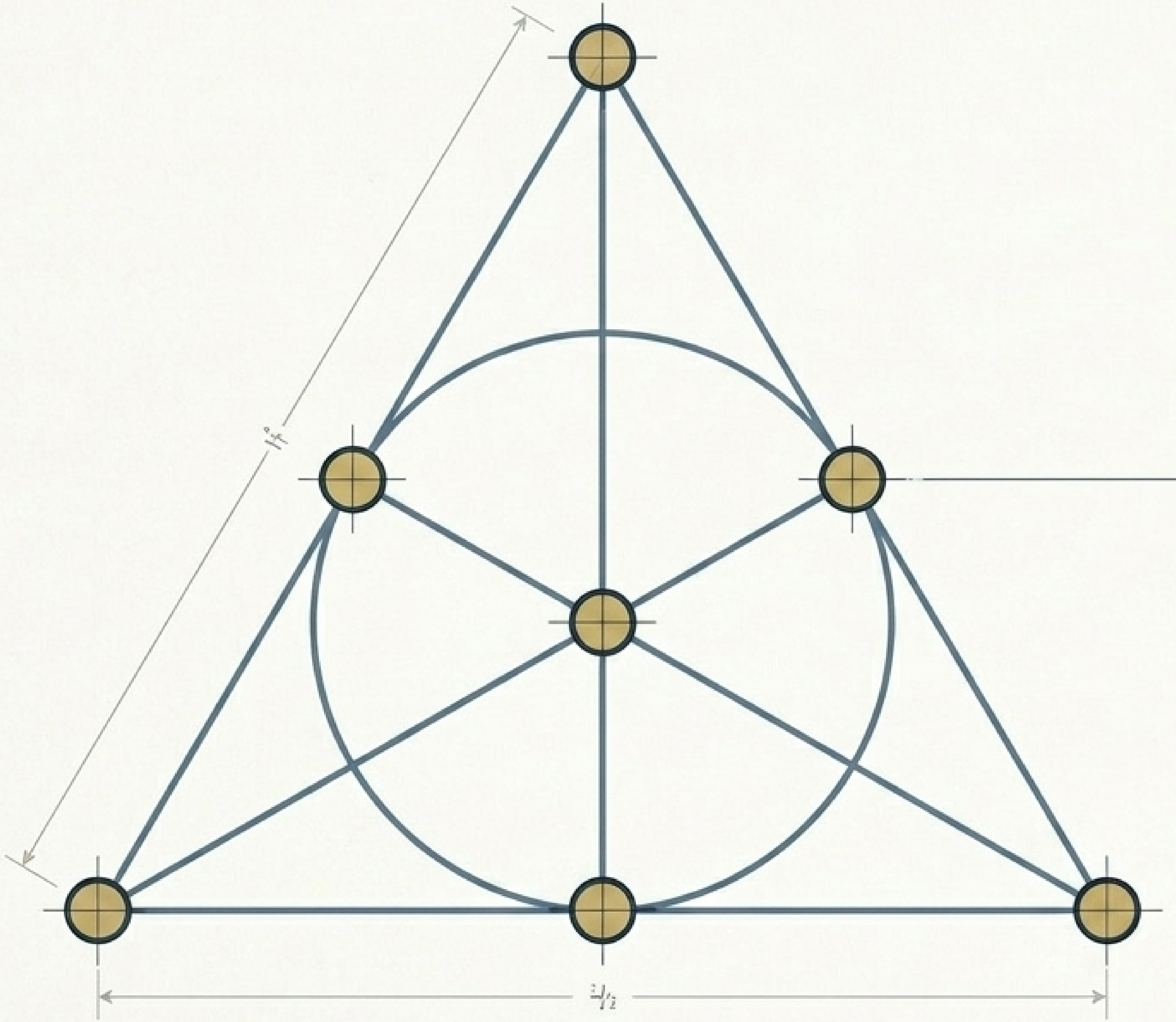
Sum of the weight block (W) = 36.

$36 = \frac{6!}{20}$.

This constant defines the geometric times cadence (the Chronograph stride) as the frame = chronograph function.

This constant defines the geometric time cadence (the Chronograph stride) as the frame moves through its 8 bounded segments.

FINITE PROJECTIVE GEOMETRY: THE FANO INTERCEPT



CORE CONCEPT

Gate 2 of the kernel validation pipeline relies on finite projective geometry to guarantee loop termination.

THE GEOMETRY OF PG(2,2):

7 total points, 7 total lines.

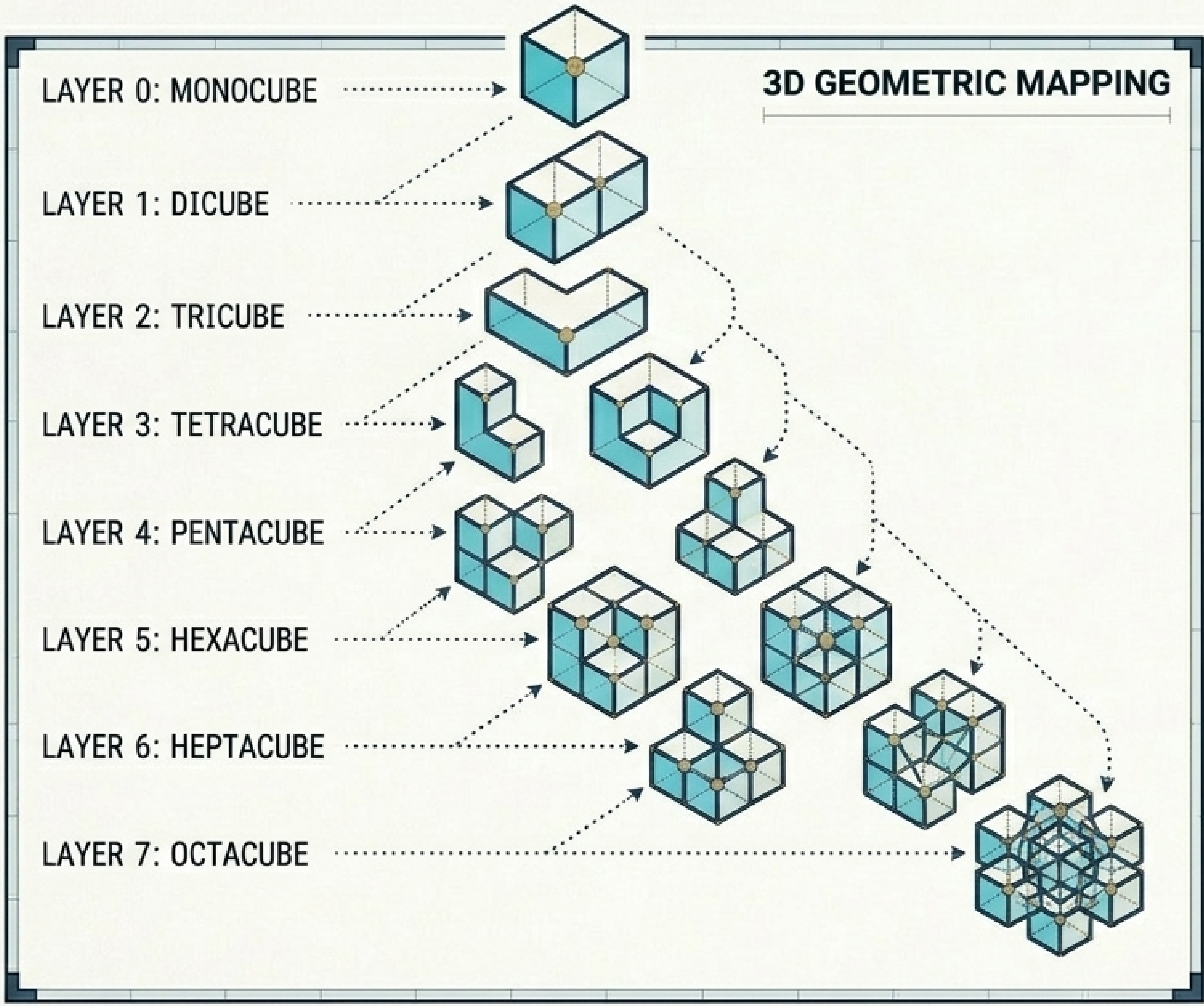
Exactly 3 points on every line, and 3 lines through every point.

THE VALIDATION GUARANTEE (TRANSYLVANIA LOTTERY):

Any Delta Law trajectory mapped through this specific projective geometry is mathematically guaranteed to intersect and resolve within a strict bound of ≤ 15 steps. If it exceeds 15, the frame is structurally invalid and dropped.

STRUCTURAL STRUCTURAL DIMENSIONS: POLYCUBES & SYMMETRY

3D GEOMETRIC MAPPING



POLYCUBE COMBINATORICS MAPPED TO OMI FACTORIAL LAYERS:

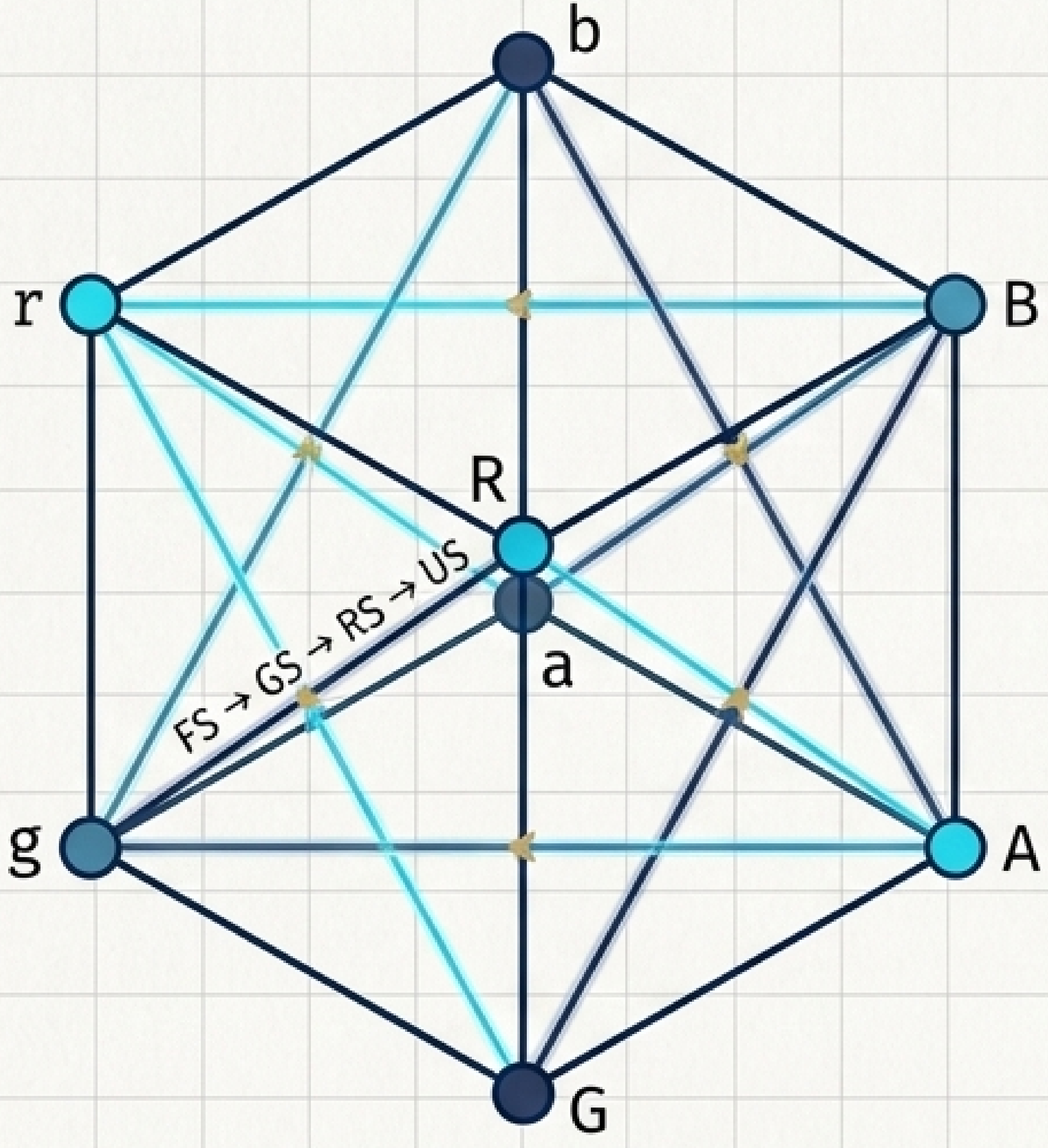
LAYER 0 (FIXED POINT)	Monocube	1 Free 1 One-Sided
LAYER 1 (MONADIC)	Dicube	1 Free 1 One-Sided
LAYER 2 (DYADIC)	Tricubes	2 Free 2 One-Sided
LAYER 3 (TRIADIC)	Tetracubes	5 Free 8 One-Sided
LAYER 4 (TETRADIC)	Pentacubes	12 Free 18 One-Sided
LAYER 5 (PENTADIC)	Hexacubes	35 Free 60 One-Sided
LAYER 6 (HEXADIC)	Heptacubes	108 Free 196 One-Sided
LAYER 7 (REPLAY RING)	Octacube	3,811 Free 6,922 One-Sided

SYMMETRY SUBGROUPS CONTEXT:

Polycubes distinguish mirror reflections (chiral pairs). Omi encodes all 33 distinct symmetry types (subgroups of the achiral octahedral group) directly onto the 16-bit segment[6] address space.

THE GEOMETRY OF LIGHT: MIQUEL-STYLE CONFIGURATIONS

THE OMI-GNOMON FORMULA: G(aA) = rRgGbBaA

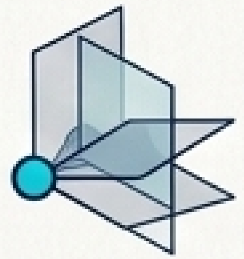


INCIDENCE STRUCTURE VISUALIZATION

INCIDENCE BREAKDOWN (THE 8_3, 6_4 STRUCTURE):

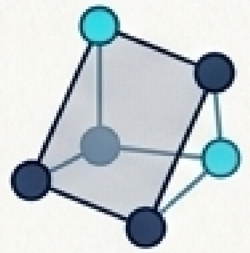
8 CHIRAL VERTICES:

- Representing light tokens (r, R, g, G, b, B, a, A).
- Each vertex participates in exactly 3 planes.



6 RELATIONAL PLANES:

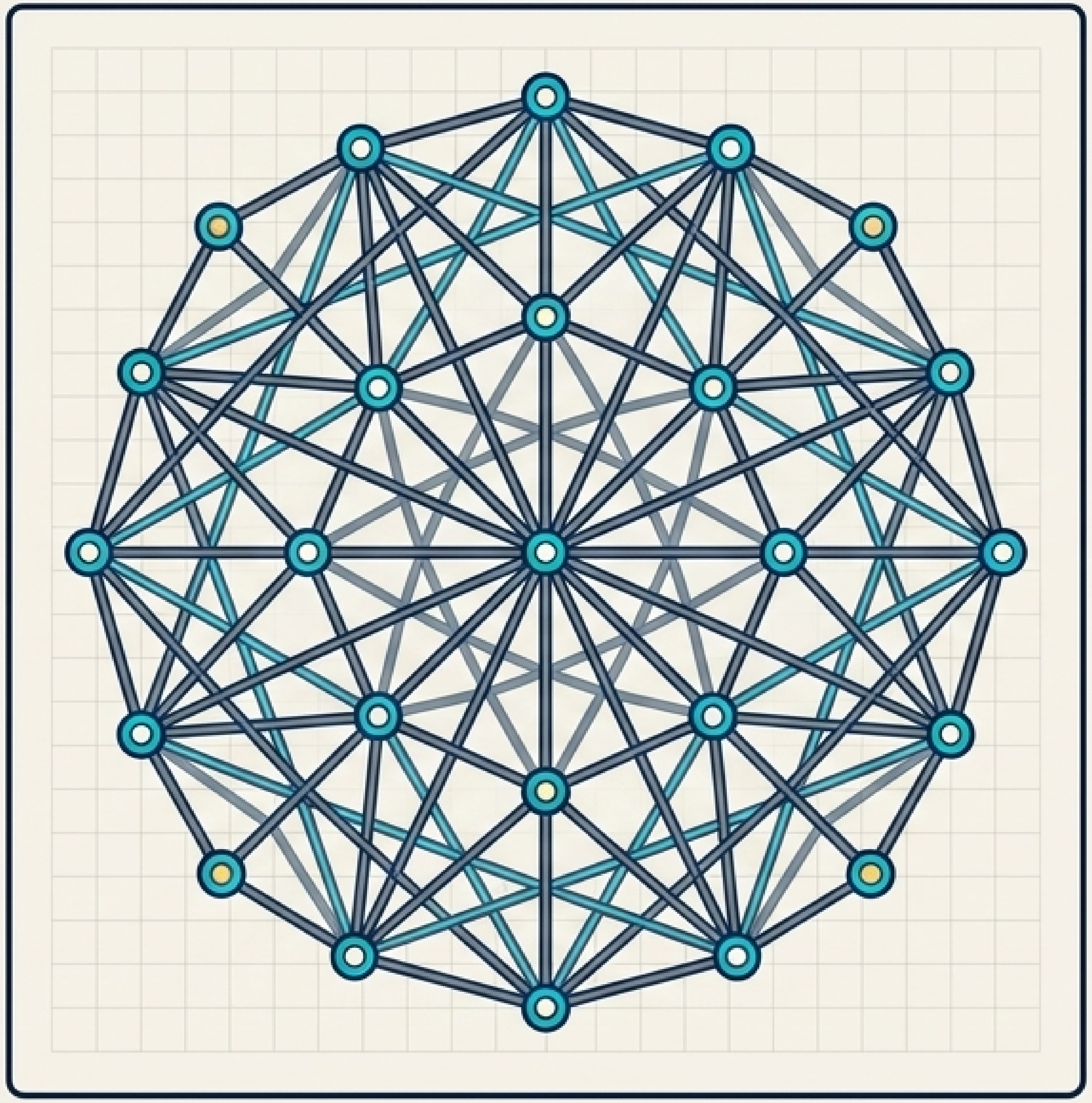
- Representing paired gates (FS-GS, FS-RS, etc.).
- Each plane contains exactly 4 vertices.



MATHEMATICAL SYNTHESIS:

This configuration acts as a pre-spacetime incidence frame. It is not an object in the world; it is the mathematical aperture through which Omi-World becomes orientable.

EXPANSIVE COMBINATORICS: THE CAYLEY-SALMON LENS



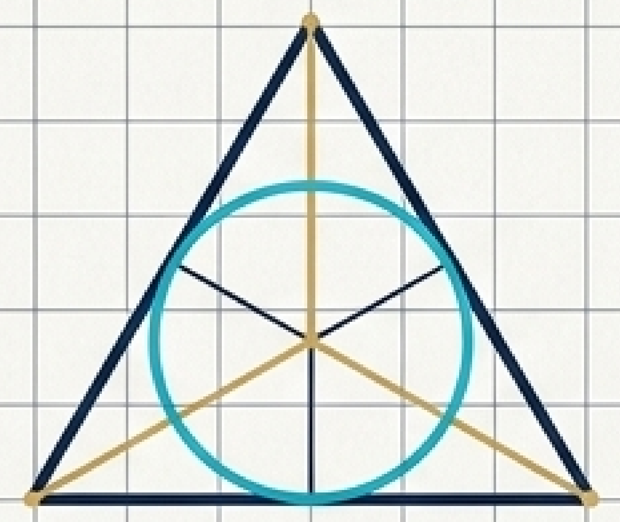
THE 15_4, 20_3 CONFIGURATION:

- **15 Points:** Representing the nonzero states of a 4-bit field (2^4 - 1). Every point has exactly 4 intersecting lines.
- **20 Lines:** Representing triadic relations/traversal paths. Every line contains exactly 3 points.

SYSTEM ROLE:

Expanding the base Desargues agreement geometry into the Omi-Psi field. It mathematically accounts for open-world observer permutations (Focus, Attention, Observation, Experience) while keeping the system finite and closed.

THE GEOMETRIC SCALING MATRIX

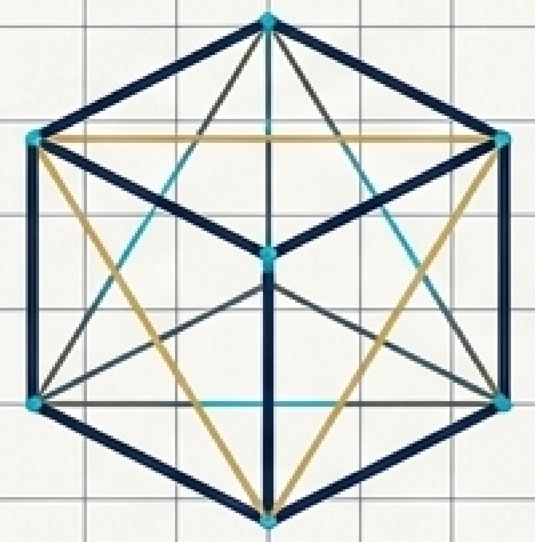


FANO PLANE

STRUCTURE:
PG(2,2) / 7 POINTS, 7 LINES.

PROTOCOL ROLE:
REPLAY, LOOP VALIDATION, RESOLUTION LIMITS.

MATHEMATICAL BOUND:
<= 15 DETERMINISTIC STEPS.

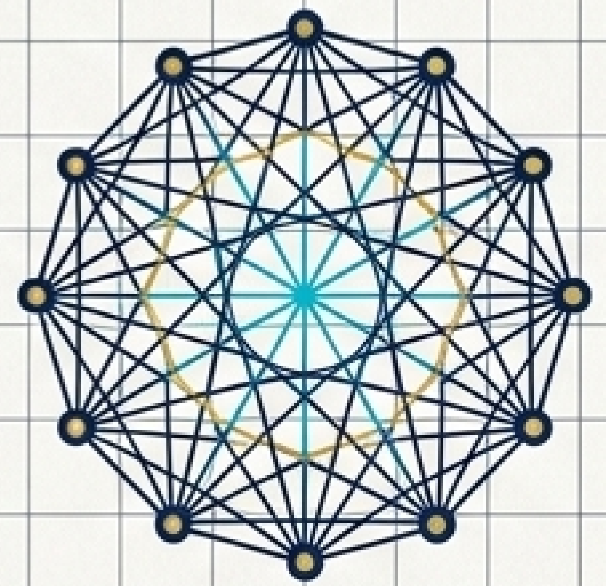


MIQUEL CONFIGURATION

STRUCTURE:
INCIDENCE (8_3, 6_4) / 8 POINTS, 6 PLANES.

PROTOCOL ROLE:
OPTICAL ORIENTATION, CUBE LIMITS, PRE-SPACETIME LIGHT-FIELD.

MATHEMATICAL BOUND:
8! LOCAL GNOMON CHANNEL PERMUTATIONS.



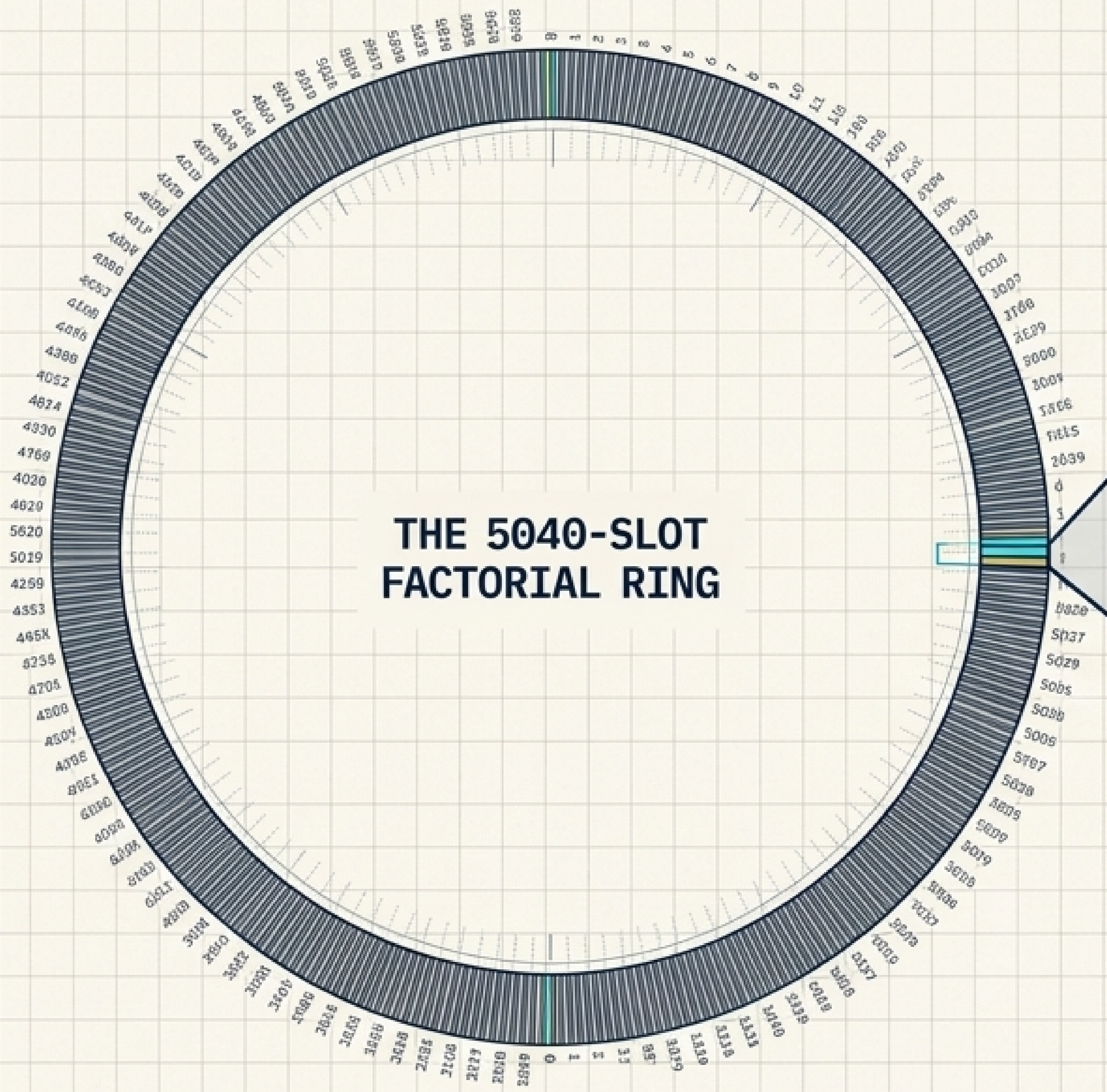
CAYLEY-SALMON

STRUCTURE:
(15_4, 20_3) / 15 POINTS, 20 LINES.

PROTOCOL ROLE:
OBSERVER STATES, COMPASS PERMUTATIONS, OMI-PSI FIELD EXPANSION.

MATHEMATICAL BOUND:
15 NONZERO STATES OF A 4-BIT FIELD.

COMBINATORIAL MEMORY: THE 5040-SLOT RING

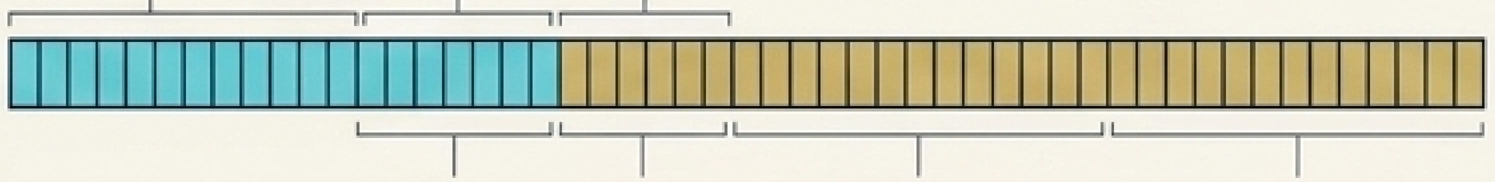


THE 7! MATHEMATICAL LAW:

Omi memory is a fixed, lock-free circular buffer containing exactly 5040 slots (7!). It advances geometrically, not chronologically.

THE ATOMIC 64-BIT BIGINT LEDGER (SLOT ANATOMY):

16 BITS: PROVENANCE (EPOCH + SUB-EPOCH TAG).
8 BITS: STEPS (DISTANCE TAKEN IN FANO RESOLUTION).



8 BITS: LL (FANO PLANE POINT IDENTIFIER 1-7).
16 BITS: NN (FREE VARIABLE / INPUT COLUMN).
16 BITS: MM (FREE VARIABLE / OUTPUT COLUMN).

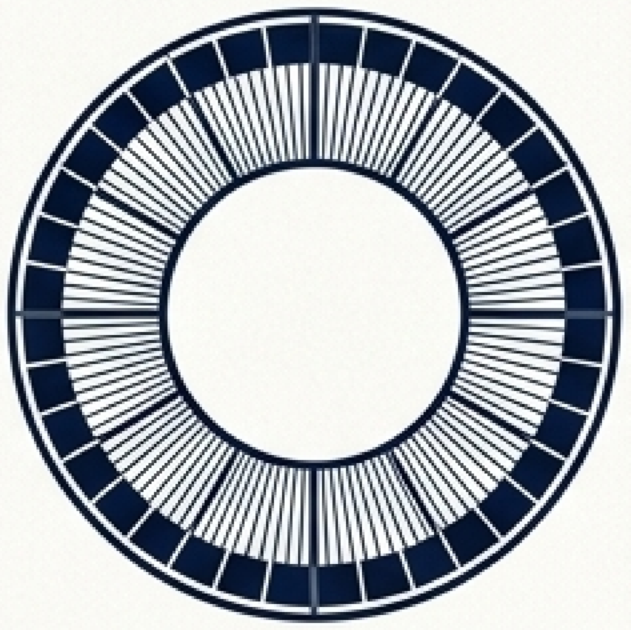
PHYSICS OF THE RING:

Memory advancement relies on an atomic compareExchange cursor. When slot 5039 is breached, the epoch modulo counter wraps around flawlessly.

SYNTHESIS: THE UNIFIED PHYSICS OF OMI

1. VALIDATE
$Q_{frame} = 0$ Fano Filter
(Image 1)

2. RESOLVE
$\delta_C(x)$
Delta Law Trajectory



3. RECORD
7! (5040) Combinatorial Ring
(Image 2)

4. PROJECT & INSPECT
Q_{xy} & G(aA) Miquel Light-Field
(Image 1)

1. Validation:

The Quadratic/Fano Filter. $Q_{frame} = 0$ blocks structural corruption at bare-metal speed.

2. Resolution:

The Delta Law ($\delta_C(x)$) computes the trajectory.

3. Replay (Memory):

The 7! (5040) Combinatorial Ring records the exact ledger state.

4. Projection:

The Spatial Equation Q_{xy} and the Miquel light-field G(aA) render the object to the observer.

FINAL TAKEAWAY: OMI IS NOT MERELY A **CODEBASE.** IT IS A STRICTLY BOUNDED, HIGHLY COMPOSITE MATHEMATICAL UNIVERSE WHERE EVERY ACTION IS AN EXACT, INVERTIBLE, AND REPLAYABLE EQUATION.